Obviously, for both equations to be true, λ must equal λ^*. In other words, λ (and therefore any eigenvalue of a Hermitian operator) must be real.

3.1.5 Hermitian Operators and Orthonormal Bases

We come now to the basic mathematical theorem—I will call it the *fundamental theorem*—that serves as a foundation of quantum mechanics. The basic idea is that *observable quantities in quantum mechanics are represented by Hermitian operators*. It's a very simple theorem, but it's an extremely important one. We can state it more precisely as follows:

The Fundamental Theorem

- The eigenvectors of a Hermitian operator are a complete set. This means that **any vector the operator can generate** can be expanded as a sum of its eigenvectors.

- If λ_1 and λ_2 are two unequal eigenvalues of a Hermitian operator, then the corresponding eigenvectors are orthogonal.

- Even if the two eigenvalues are equal, the corresponding eigenvectors can be chosen to be orthogonal. This situation, where two different eigenvectors have the same eigenvalue, has a name: it's called *degeneracy*. Degeneracy comes into play when two operators have simultaneous eigenvectors, as discussed later on in Section 5.1.

One can summarize the fundamental theorem as follows: The eigenvectors of a Hermitian operator form an orthonormal basis. Let's prove it, beginning with the second bullet item.

According to the definition of eigenvectors and eigenvalues, we can write